Appendix 3. Countable and uncountable sets

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A3.1 Cardinality and bijections

Write $A \sim B$ if a bijection $A \to B$ exists. Cardinality abstracts this equal-size relation. An infinite set can be equinumerous with a proper subset: $n \mapsto 2n$ bijects $\mathbb{N}^+ = \{1, 2, \ldots\}$ with $2\mathbb{N}^+$. Equal cardinalities do not make these sets equal.

For finite sets, element counts compare size. Infinite cardinality needs existence of bijections rather than the finite intuition that every proper subset is smaller.

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A3.2 Countable and countably infinite

A set is finite if it bijects with $\{1, \ldots, n\}$, with the empty set assigned size zero, and infinite otherwise. Countable means finite or equinumerous with $\mathbb{N}^+$. An infinite countable set is countably infinite; uncountable means not countable.

A countable set has an enumeration $A = \{a_1, a_2, \ldots\}$, permitting repetition or termination for finite sets. Equivalently, a nonempty countable set is the image of a surjection from $\mathbb{N}^+$. Treat the empty set separately, since no function maps a nonempty domain to it. For infinite images, remove duplicates to obtain distinct representatives and a bijection. A sequence may repeat terms; a distinct-element listing requires duplicate removal.

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A3.3 Integers, pairs, and pairing

The listing $0, 1, -1, 2, -2, \ldots$ enumerates integers, with $f(1) = 0$, $f(2n) = n$, $f(2n+1) = -n$. Starting natural numbers at zero, Cantor pairing is

$$\pi(m, n) = \frac{(m+n)(m+n+1)}{2} + m.$$

It bijects $\mathbb{N}_0^2$ with $\mathbb{N}_0$, following diagonals

$$(0,0);\ (0,1),(1,0);\ (0,2),(1,1),(2,0);\ldots.$$

A listing needs a finite index for every object, not a uniquely natural ordering.

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A3.4 Closure properties

Subsets of a countable set are countable by retaining their terms in a listing. Products with $\mathbb{N}_0$ and finite products of countable sets are countable by repeated pairing. Countably infinite unions require the conditions on the next section.

Every rational is p/q for $p \in \mathbb{Z}, q \in \mathbb{N}^+$, without requiring reduced form. The map $(p, q) \mapsto p/q$ surjects from a countable product onto $\mathbb{Q}$, including zero via $p = 0$. Remove repeated values to enumerate rationals. Distinct fractions such as $1/2, 2/4$ may represent the same value, so the first map is a surjection.

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A3.5 Finite and countably infinite unions

For finitely many countable sets, first use the tagged disjoint union

$$\bigsqcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} (A_i \times \{i\}).$$

Finite interleaving lists it. Projection $(a, i) \mapsto a$ has image the original union, proving it countable. Original elements are not tagged pairs, so the original union is not automatically a subset of this tagged union. Skip empty sets.

For a countably indexed family with simultaneously specified enumerations $A_n = \{a_{n,1}, a_{n,2}, \ldots\}$, the map $(n, k) \mapsto a_{n,k}$ from a countable pair domain enumerates the union.

If only the separate countability of each A_n is known, simultaneously selecting their enumerations is a choice issue in ZF. Countable Choice suffices; the default background here is standard ZFC unless stated otherwise. A finite family has no infinite simultaneous-choice gap. For an infinite family, uniform enumeration or the relevant choice assumption is part of the proof's conditions.

A3.6 Diagonalizing subsets of natural numbers

Using positive natural numbers, suppose all subsets were listed as $S_1, S_2, \ldots$. Define

$$D = \{n \in \mathbb{N}^+ : n \notin S_n\}.$$

For each k, membership of k in D is opposite to its membership in S_k. Hence $D \neq S_k$ for every k, contradicting completeness of the list.

More generally, Cantor's theorem gives $|A| < |\mathcal{P}(A)|$. Finite sets are countable, but powersets of infinite sets need not be. This proof uses membership directly, without decimal conventions.

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A3.7 Uncountability and decimal ambiguity

If $(0, 1]$ were countable, its subset $(0, 1)$ would be. Remove endpoint one, and list the remaining values using canonical decimals $r_n = 0.a_{n1}a_{n2}\cdots$, excluding eventually-all-nine alternatives. Define

$$b_k = \begin{cases} 3, & a_{kk} \neq 3, \\ 7, & a_{kk} = 3, \end{cases} \qquad x = 0.b_1 b_2 \cdots .$$

Digits three and seven ensure $0 < x < 1$ with no trailing-nine ambiguity. The k-th digit differs from that of r_k, so x is missing, contradiction. Thus $(0, 1)$, its superset $(0, 1]$, and $\mathbb{R}$ are uncountable.

Without resolving $0.1000\cdots = 0.0999\cdots$, a digit mismatch might concern representations rather than distinct numbers.

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A3.8 Directions of countability arguments

Subsets of countable sets are countable. Supersets of uncountable sets are uncountable, but their every subset need not be; finite subsets still exist. Since $\mathbb{Q}$ is countable and $\mathbb{R}$ uncountable, the irrationals are uncountable. Otherwise

$$\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$$

would be a union of two countable sets. Infinite cardinal sums and products need bijection arguments rather than unexamined finite arithmetic. State the sets, function direction, and any choice assumptions in each countability claim.

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A3.9 Reconstructing the pairing inverse

Let $T_s = s(s+1)/2$, so $\pi(m,n) = T_{m+n} + m$. Diagonal s has $s+1$ pairs and exactly the integer indices in $[T_s, T_{s+1})$.

Given $z \geq 0$, there is a unique s with $T_s \leq z < T_{s+1}$. Put $m = z - T_s, n = s - m$. Interval length gives $0 \leq m \leq s$, so both coordinates are nonnegative and $\pi(m,n) = z$. Every component is unique, proving both onto and one-to-one.

Mathematically $s = \lfloor (\sqrt{8z+1} - 1)/2 \rfloor$. Large-integer code should use integer square root or comparisons instead of rounding floating-point approximations.

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A3.9 Reconstructing the pairing inverse (continued)

Cantor pairing: row m, column n; cells contain π(m,n)

	n=0	n=1	n=2	n=3
m=0	0	1	3	6
m=1	2	4	7	11
m=2	5	8	12	17
m=3	9	13	18	24

Green diagonal: m+n=3

π=6,7,8,9 (upper right → lower left)

Invert 18: $15=T_5 \le 18 < T_6=21$

m=18−15=3 , n=5−3=2

Window: 0≤m,n≤3. Complete diagonals occupy consecutive, disjoint index intervals.

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A3.10 Removing duplicates and listing subsets

For a surjection $f : \mathbb{N}_0 \to A$, take first-occurrence indices

$$I = \{n : \forall k < n, f(k) \neq f(n)\}.$$

The restriction $f|_I$ bijects I with A. Finite I gives finite A. If I is infinite, define $i_0 = \min I$, then $i_{t+1} = \min\{i \in I : i > i_t\}$. Each step exists. Every $i \in I$ has only finitely many smaller natural numbers, so it is eventually listed. Thus $I \sim \mathbb{N}_0$ and A is countable.

For $B \subseteq A$, use a distinct listing of A and retain its indices belonging to B. The same finite-or-infinite argument applies; empty B is handled separately. This proves set existence, not a computable membership test for arbitrary B.

With given union enumerations, map tagged pairs to elements, sort indices by pairing, and remove duplicates. Skip empty A_n instead of inventing enumerations for them.

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A3.11 Finite words, finite subsets, and infinite sequences

For a finite alphabet Σ of size k, words of length n number k^n. Length-first, then lexicographic order lists Σ^*, proving countability. It is infinite for $k \geq 1$; for $k = 0$, only the empty word remains.

Finite subsets of $\mathbb{N}_0$ have an explicit bijection with $\mathbb{N}_0$:

$$S \longmapsto \sum_{i \in S} 2^i.$$

Uniqueness of binary expansion proves injectivity, and every integer's finite bit support gives the inverse, with zero corresponding to the empty set.

By contrast, $\{0,1\}^{\mathbb{N}_0}$ is uncountable. Against any listing $s_0, s_1, \ldots$, define $t_i = 1 - s_i(i)$, differing from row i at coordinate i. The infinite product is not the union of finite-length products. Countability of all finite-length layers does not prove countability of infinite sequences.

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A3.12 Cantor's theorem for every set

Suppose $f : A \to \mathcal{P}(A)$ is onto and define

$$D = \{a \in A : a \notin f(a)\}.$$

Onto supplies d with $f(d) = D$, giving $d \in D \iff d \notin D$, contradiction. No such surjection or bijection exists. The singleton map $a \mapsto \{a\}$ is injective, hence $|A| < |\mathcal{P}(A)|$.

For empty A, its powerset is the nonempty singleton $\{\varnothing\}$, and no surjection exists either. The proof needs neither decimal representations nor a particular listing order.

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A3.12 Cantor's theorem for every set (continued)

Given a proposed list $S_0, S_1, \ldots$, reverse coordinate i of row i to construct D

	i=0	i=1	i=2	i=3
S0	1	0	1	1
S1	0	0	1	0
S2	1	1	1	0
S3	0	1	0	0

First four digits of D: 0,1,0,1

$D(i) = 1 - 1[i \in S_i]$, defined for every i.

D differs from S_0 at coordinate 0,

from S_1 at coordinate 1, and so on.

If D=S_m, coordinate m contradicts it.

The figure shows four coordinates to illustrate reversal. The formal definition covers every index. A finite picture does not claim a finite set has infinite cardinality.

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A3.13 Constructing Cantor–Schröder–Bernstein

Given injections $f : A \to B, g : B \to A$, define

$$A_0 = A \setminus g(B), \quad A_{n+1} = g(f(A_n)), \quad C = \bigcup_{n \geq 0} A_n,$$

$$h(a) = \begin{cases} f(a), & a \in C, \\ g^{-1}(a), & a \notin C. \end{cases}$$

The second branch is defined because $a \notin A_0$ implies $a \in g(B)$, with unique preimage under the injection. Each branch is injective. If a cross-branch equality $f(c) = g^{-1}(a)$ occurred for $c \in C, a \notin C$, then $a = g(f(c)) \in C$, contradiction.

For arbitrary $b \in B$, if $g(b) \notin C$, then $h(g(b)) = b$. Otherwise $g(b) \in C$ but not A_0, so it lies in some $A_{n+1} = g(f(A_n))$. Injectivity of g gives $b = f(c) = h(c)$ for some $c \in A_n \subseteq C$. Thus h is onto and a bijection. No arbitrary selection of preimages is needed; injection determines each uniquely.

A3.14 Reals and the powerset of natural numbers

Inject $\mathcal{P}(\mathbb{N}_0)$ into $[0,1] \subseteq \mathbb{R}$ by

$$S \longmapsto \sum_{i \geq 0} \frac{\mathbf{21}_{i \in S}}{3^{i+1}}.$$

If two sets first differ at k, that digit contributes difference $2/3^{k+1}$. The entire later tail can offset at most $1/3^{k+1}$, leaving a positive difference. This proves injectivity, including ternary boundary cases.

Conversely, fix a distinct rational enumeration (q_i), and send x to $\{i : q_i < x\}$. If $x < y$, rational density supplies a q_i between them, distinguishing their sets. To prove density, choose integer N with $N(y-x) > 1$, then integer $Nx < k < Ny$; k/N lies between x, y. CSB gives

$$|\mathbb{R}| = |\mathcal{P}(\mathbb{N}_0)| = 2^{\aleph_0}.$$

Every nonempty open real interval has this cardinality. Map linearly to $(-1,1)$, then use $h(t) = t/(1-|t|)$, whose inverse is $y/(1+|y|)$. This is an explicit bijection rather than an appeal to density.

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A3.15 Density, cardinality, and computability

Rationals are countable yet meet every nonempty open interval; density differs from cardinality. Real algebraic numbers are countable. Integer polynomials are finite integer tuples, hence countable, and a nonzero degree-d polynomial has at most d roots by induction using the factor theorem. A countable union of their finite real-root sets is countable. If transcendental reals were also countable, their union with algebraic reals would make all reals countable, contradiction.

Finite programs over a fixed finite alphabet are countable. Each program computes at most one total binary function, so computable total binary functions are countable. All functions $\mathbb{N}_0 \to \{0, 1\}$ biject with $\mathcal{P}(\mathbb{N}_0)$ and are uncountable. Thus noncomputable binary functions exist. This existence proof gives no algorithm deciding whether a specified program halts.

If A is uncountable and C countable, then $A \setminus C$ is uncountable. Otherwise $A = (A \setminus C) \cup (A \cap C)$ would be a union of two countable sets. It does not follow that every subset is uncountable.

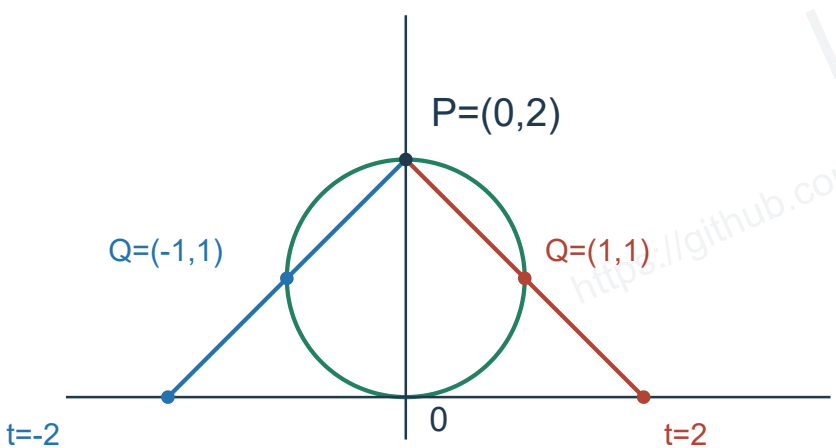
A3.16 A projection between a line and a punctured circle

Let C be $x^2 + (y-1)^2 = 1$, with $P = (0, 2)$. The line from P to $(t, 0)$ meets the circle again at

$$\phi(t) = \left(\frac{4t}{t^2+4}, \frac{2t^2}{t^2+4} \right).$$

A line point is $(st, 2-2s)$. Substitution in the circle equation gives $s^2(t^2+4) - 4s = 0$. Root zero gives P; the other root $4/(t^2+4)$ gives ϕ. At $t = 0$, it still gives $(0, 0)$, without exception.

Projection from P=(0,2): each second circle intersection corresponds to one real t



Circle: x²+(y−1)²=1; P is excluded.

φ(t)=(4t/(t²+4), 2t²/(t²+4))

inverse : t=2x/(2−y) · Q≠P。

t=0 uses the y-axis and maps to (0,0).

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A3.16 A projection between a line and a punctured circle (continued)

For $Q = (x, y) \in C \setminus \{P\}$, $y < 2$, and the inverse is $t = 2x/(2 - y)$. Substituting $x^2 = 2y - y^2$ verifies $\phi(t) = Q$, proving onto and one-to-one.

Adding P back does not change cardinality, as an explicit bijection sends $\mathbb{R}$ plus a new element $*$ to $\mathbb{R}$: send $*$ to zero, each nonnegative integer n to $n + 1$, and leave every other real fixed. Combining with ϕ bijects the full circle with the reals. The reason is the construction, rather than treating "adding one changes nothing" as a finite-arithmetic rule.

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References and attribution

Ralph P. Grimaldi. Discrete and Combinatorial Mathematics: An Applied Introduction, 5th Edition, 2004.

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